Year 9
Mathematics

Unit 3: Sequences and Algebra

3.3 Expanding

Expanding

STARTER · 5 MIN

Date: _____ **Title: Expanding**

LEARNING OBJECTIVES

- Expand two brackets
- Square a bracket and expand $(x + a)(x - a)$
- Use expanding to show identities

STARTER *Answer in your book. No calculators.*

1. Expand $3(x + 4)$
2. Expand $x(x + 5)$
3. Work out -4×5
4. Work out -3×-6

Starter — Answers

STARTER

Tick or correct in green pen

1. $3x + 12$

2. $x^2 + 5x$

3. -20

4. 18

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can expand two brackets	☐
SILVER	I can square a bracket and expand $(x + a)(x - a)$	☐
GOLD	I can use expanding to show identities	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Expanding

INTRODUCTION · 3 MIN

Multiply every term in one bracket by every term in the other.

Two brackets

Four multiplications, then collect.

$(x + 2)(x + 5) = x^2 + 7x + 10$

Signs

$- \times + = -$ and $- \times - = +$

$(x - 4)(x - 2) = x^2 - 6x + 8$

Squaring

Write the bracket twice.

$(x + 3)^2 = (x + 3)(x + 3)$

Difference of squares

The x terms cancel.

$(x + 5)(x - 5) = x^2 - 25$

Examples 1 and 2 — Expanding Two Brackets

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint multiply every term by every term
− × − = + − × + = −

Example 1

Expand and simplify

$(x + 2)(x + 5)$

Example 2

Expand and simplify

(a) $(x + 3)(x - 5)$

(b) $(x - 4)(x - 2)$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

x × x, x × 5, 2 × x, 2 × 5

$x^2 + 5x + 2x + 10$

$= x^2 + 7x + 10$

Tip: collect the two x terms.

Example 2

Watch the signs

(a) $x^2 - 5x + 3x - 15$

$= x^2 - 2x - 15$

(b) $x^2 - 2x - 4x + 8$

$= x^2 - 6x + 8$

Tip: $-4 \times -2 = +8$

Task 1

PRACTICE · 9 MIN

BRONZE *Expand and simplify*

Hint multiply every term by every term − × − = + − × + = −

1 $(x + 3)(x + 4)$

2 $(x + 1)(x + 6)$

3 $(x + 5)(x - 2)$

4 $(x - 3)(x + 7)$

5 $(x - 4)(x - 5)$

★ $(x - 8)(x - 1)$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x^2 + 7x + 12$

2 $x^2 + 7x + 6$

3 $x^2 + 3x - 10$

4 $x^2 + 4x - 21$

5 $x^2 - 9x + 20$

★ $x^2 - 9x + 8$

Examples 3 and 4 — Squares and Differences of Squares

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $(x + a)^2 = (x + a)(x + a)$ $(x + a)(x - a) = x^2 - a^2$

Example 3

Expand and simplify

(a) $(x + 5)^2$

(b) $(x - 3)^2$

Example 4

Expand and simplify

$(x + 6)(x - 6)$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Write the bracket twice

(a) $(x + 5)(x + 5)$

$= x^2 + 10x + 25$

(b) $(x - 3)(x - 3)$

$= x^2 - 6x + 9$

Tip: $(x + 5)^2$ is not $x^2 + 25$.

Example 4

The x terms cancel

$x^2 - 6x + 6x - 36$

$= x^2 - 36$

Tip: $(x + a)(x - a) = x^2 - a^2$

Task 2

PRACTICE · 9 MIN

SILVER *Expand and simplify*

Hint $(x + a)^2 = (x + a)(x + a)$ $(x + a)(x - a) = x^2 - a^2$

1. $(x + 4)^2$
2. $(x + 1)^2$
3. $(x - 6)^2$
4. $(x + 2)(x - 2)$
5. $(x - 10)(x + 10)$

★ $(x - 11)^2$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x^2 + 8x + 16$
2. $x^2 + 2x + 1$
3. $x^2 - 12x + 36$
4. $x^2 - 4$
5. $x^2 - 100$

★ $x^2 - 22x + 121$

Examples 5 and 6 — Identities and Area

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint expand, then collect like terms area = length × width

Example 5

(3 marks)

Show that

$(x + 3)(x + 5) - (x + 2)^2$

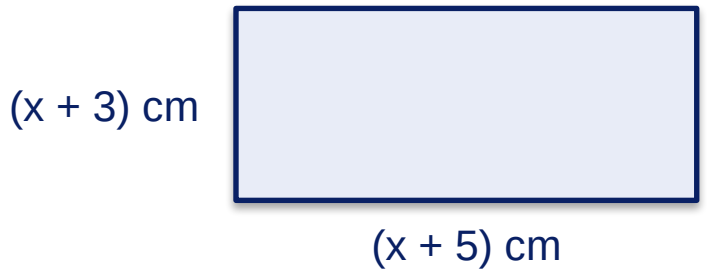
$\equiv 4x + 11$

Example 6

(3 marks)

The area of the rectangle is 35 cm^2. Show that

$x^2 + 8x - 20 = 0$



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Expand both, then subtract

$x^2 + 8x + 15 - (x^2 + 4x + 4)$

$= x^2 + 8x + 15 - x^2 - 4x - 4$

$\mathbf{= 4x + 11}$

Tip: brackets round the part you subtract.

Example 6

Area = length × width

$(x + 5)(x + 3) = 35$

$x^2 + 8x + 15 = 35$

$\mathbf{x^2 + 8x - 20 = 0}$

Tip: subtract 35 from both sides.

Task 3

PRACTICE · 9 MIN

GOLD *Show your working*

Hint expand, then collect like terms area = length × width

1 Show that
$(x + 1)(x + 4) - x^2 \equiv 5x + 4$

2 Show that
$(x + 2)^2 - (x - 2)^2 \equiv 8x$

3 Expand and simplify
$(x + 3)^2 + (x - 1)^2$

4 Rectangle $(x + 6)$ by $(x + 1)$, area 50.
Show that $x^2 + 7x - 44 = 0$

5 Square with side $(x + 3)$, area 64.
Show that $x^2 + 6x - 55 = 0$

★ Parallelogram: base $(x + 7)$, height $(x + 2)$, area 60. Show $x^2 + 9x - 46 = 0$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $x^2 + 5x + 4 - x^2 = 5x + 4$

2 $x^2 + 4x + 4 - (x^2 - 4x + 4) = 8x$

3 $2x^2 + 4x + 10$

4 $x^2 + 7x + 6 = 50$

so $x^2 + 7x - 44 = 0$

5 $x^2 + 6x + 9 = 64$

so $x^2 + 6x - 55 = 0$

★ $x^2 + 9x + 14 = 60$

so $x^2 + 9x - 46 = 0$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Expand and simplify $(x + 3)(x + 2)$

SILVER Expand and simplify $(x - 4)^2$

GOLD Show that $(x + 1)^2 - x^2 \equiv 2x + 1$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x^2 + 5x + 6$

SILVER $x^2 - 8x + 16$

GOLD $x^2 + 2x + 1 - x^2 = 2x + 1$

Exam Practice — Two Brackets

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint multiply every term by every term
− × − = + − × + = −

Question 1

(4 marks)

Expand the brackets and simplify.

(a) (x + 2)(x + 6) (2)

.....

(b) (x − 3)(x − 5) (2)

.....

Question 2

(2 marks)

Expand and simplify
(x + 4)(x − 9)

.....

Exam Practice — Squares

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint

$(x + a)^2 = (x + a)(x + a)$

$(x + a)(x - a) = x^2 - a^2$

Question 3

(4 marks)

Expand and simplify.

(a) $(x + 8)^2$ (2)

.....

(b) $(x - 5)(x + 5)$ (2)

.....

Question 4

(2 marks)

Ali says $(x + 4)^2 = x^2 + 16$.

Is he correct? Explain your answer.

.....

Exam Practice — Identities and Area

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint expand, then collect like terms
area = length × width

Question 5

(3 marks)

Show that

$(x + 4)^2 - (x + 1)(x + 7) \equiv 9$

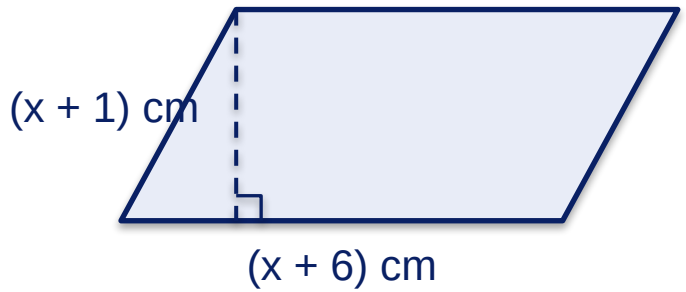
Question 6

(3 marks)

The parallelogram has area 30 cm^2.

Show that $x^2 + 7x - 24 = 0$

Diagram NOT accurately drawn



Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $x^2 + 6x + 2x + 12$ — M1

$= x^2 + 8x + 12$ — A1

(b) $x^2 - 5x - 3x + 15$ — M1

$= x^2 - 8x + 15$ — A1

Tip: − × − = +

Question 2

$x^2 - 9x + 4x - 36$ — M1

$= x^2 - 5x - 36$ — A1

Tip: −9x + 4x = −5x

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $x^2 + 8x + 8x + 64$ — M1

$= x^2 + 16x + 64$ — A1

(b) $x^2 + 5x - 5x - 25$ — M1

$= x^2 - 25$ — A1

Tip: (b) the x terms cancel.

Question 4

$(x + 4)(x + 4) = x^2 + 8x + 16$ — M1

No: he missed the 8x — A1

Tip: write the bracket twice.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x^2 + 8x + 16 - (x^2 + 8x + 7)$ — M1 M1

$= 9$ — A1

Tip: expand both, then subtract.

Question 6

$(x + 6)(x + 1) = 30$ — M1

$x^2 + 7x + 6 = 30$ — M1

$x^2 + 7x - 24 = 0$ — A1

Tip: area = base × vertical height